



Effects of micro/nanoplastics on oxidative damage and serum biochemical parameters in rats and mice: a meta-analysis

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Abstract Micro/nanoplastics (MNPs) are emerging as environmental pollutants with potential threats to human health. The accumulation of MNPs in the body can cause oxidative stress and increase the risk of cardiovascular disease (CVD). With the aim to systematically evaluate the extent of MNPs-induced oxidative damage and serum biochemical parameters in rats and mice, a total of 36 eligible articles were included in this meta-analysis study. The results reported that MNPs can significantly increase the levels of oxidants such as reactive oxygen species (ROS) and malondialdehyde (MDA) ($P < 0.05$), and resulted in notable increase in serum biochemical parameters including aspartate aminotransferase (AST) and alanine aminotransferase (ALT) ($P < 0.05$). Conversely, MNPs significantly reduced levels of antioxidants such as superoxide dismutase (SOD), glutathione (GSH), glutathione peroxidase (GPx) and catalase (CAT) ($P < 0.05$). Subgroup analysis revealed that

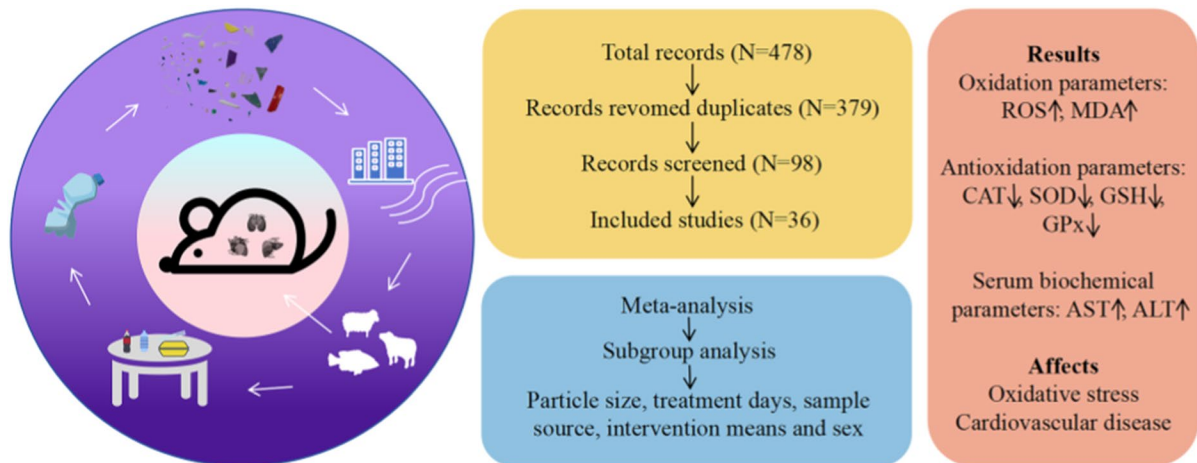
smaller MNPs with oral administration and prolonged treatment, were associated with more pronounced oxidative stress and enhanced serum biochemical parameters alteration. In addition, after affected by MNPs, the levels of ALT and AST in liver group (SMD = 2.26, 95% CI = [1.59, 2.94] and SMD = 3.10, 95% CI = [1.25, 4.94]) were higher than those in other organs. These comprehensive results provide a scientific foundation for devising strategies to prevent MNPs-induced damage, contributing to solution of this environmental and health challenge.

Graphical abstract 36 eligible articles were included in this meta-analysis and subgroup analysis was carried out based on particle size, treatment days, sample source, intervention means and sex. Micro/nanoplastics exposure increased oxidation and serum biochemical parameters, but decreased antioxidation parameters, which could induce oxidative stress and further cause cardiovascular disease.

Junliang Chen and Ruiquan Qi have contributed equally to this work.

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Introduction

Plastics are pervasive in every aspect of modern socio-economic activities, spanning food and clothing, to cosmetics and automobiles (Filiciotto & Rothenberg, 2021). Over time, plastic waste can gradually disintegrate into substantial quantities of micro/nanoplastic particles (MNPs) in the environment (Andrady, 2017). MNPs vary in size, ranging of 1 nm–5 mm and are prevalent across global ecosystem (Coffin et al., 2022; Rochman, 2018). The increasing identification of MNPs in air, food and drinking water have sparked growing concerns regarding their impacts on human health (Fang et al., 2022). Given the spreading contamination of MNPs, human exposure through digestion, inhalation, or skin contact has become inevitable (Lu et al., 2023). Since the first discovery of nine MNPs in the feces of healthy people (Schwabl et al., 2019), the relationship between MNPs and human health has been marked by a number of researchers. The detection of MNPs in human blood samples proved the MNPs' transportation through the bloodstream (Leslie et al., 2022), and the presence of MNPs in human lung tissue suggested their potential roles in the development

of lung disease (Jenner et al., 2022). To address the environmental and health issues caused by MNPs, widespread attention has been garnered, leading to calls for the establishment of relevant policies and the implementation of plastic categorization (Rochman et al., 2013a, b).

Human body encounters MNPs through three main routes: inhalation, ingestion, and skin contact (Prata et al., 2020; Rahman et al., 2021a, b), among which food consumption stands as a significant source, with approximately 39–52 k MNPs were orally taken each year and differences in age and sex could affect the intake (Cox et al., 2019; Galloway, 2015). Once ingested, MNPs can traverse the intestinal barrier, entering the circulatory system and eventually depositing in various organs including kidney, intestine, and liver (Deng et al., 2017). Inhalation of airborne MNPs provides another route of exposure, inducing lung inflammation and further severe respiratory symptoms (Atis et al., 2005; Rahman et al., 2021a, b). Notably, MNPs were discovered to possess the ability to breach the lung cell barrier and metastasize to secondary organs, changing neurotransmitter levels and functionally disrupting the nervous system (Prüst et al., 2020). Furthermore, skin contact serves as a plausible route for MNPs exposure, with particles smaller than 100 nm speculated to penetrate the skin barrier (Revel et al., 2018). Therefore, the study of MNPs and their toxicological mechanisms is of paramount importance.

Currently, a considerable body of researches on the exposure of rats/mice to MNPs were well-studied, enhancing the understanding of the MNPs toxicity by delving into the underlying mechanisms in rats/mice model. Wei et al. (2021) reported that microplastics adversely impaired heart structure and function, resulting in disruption in mitochondrial integrity. In the study of Deng et al. (2021), a significant increase in the accumulation of phthalates was observed in the liver and intestine after 30 days of exposure to phthalate-contaminated microplastics. In addition, Haddadi et al. (2022) first detected polystyrene microplastic particles in the duodenum, further identifying the effect of the exposure on ovarian dysfunction in rats. However, it's notable that there was a lack of systematic sorting of experiments on rats/mice exposure to MNPs. As a result, we intend to address this gap by exploring the MNPs exposure in rats/mice with meta-analysis.

The effects of MNPs on human health may be influenced by factors such as the characteristics of the MNPs, exposure time, dosage, etc. It has been reported that animals were primarily affected by toxicity through mechanisms such as oxidative stress and inflammation under the influence of MNPs (Sangkham et al., 2022). Researches demonstrated that exposure to MNPs was associated with histologic changes and oxidative stress induction in organisms (Rochman et al., 2013a, b). Furthermore, the detection of microplastics in human blood, suggested their potential to invade and harm the cardiovascular system (Leslie et al., 2022; Lett et al., 2021). Therefore, it's imperative to ascertain the role of oxidative stress and serum biochemical parameters in the toxicity of MNPs. Additionally, the disturbances in both oxidative and antioxidant systems following exposure to MNPs, manifested reduced CAT levels but increased MDA levels (Meng et al., 2022).

Given the absence of systematic analysis regarding the relationship between MNP-induced toxicity, serum biochemical parameters and oxidative stress, a systematic review was critically needed. This study performed a meta-analysis of experimental studies on MNP-induced oxidative damage and serum biochemical parameters in rats and mice elucidating their pivotal role in MNP-induced injuries.

Methods

Search strategy

We searched PubMed, Web of Science and Cochrane databases for eligible studies up to 16 October 2022 (Suppl Table S1) with no filter applied. All studies were searched by the following strategies: (1) nanoplastic or microplastic; (2) rats or mice; (3) inflammation or oxidative stress or toxicity; (4) terms (1) and (2) and (3).

Inclusion criteria

The inclusion criteria for meta-analysis were as follows: (1) Randomized controlled adult rat or mouse experiments, and studies published in English. (2) Regardless of strain and gender. (3) The experimental group was considered to be exposed to MNPs, the control group was a blank, and oxidative damage and serum biochemical parameters served as the result. (4) The highest dose and longest experimental period were selected for analysis. The indicators of oxidative damage and serum biochemical parameters included superoxide dismutase (SOD), malondialdehyde (MDA), reactive oxygen species (ROS), reduced glutathione (GSH), glutathione peroxidase (GPx), catalase (CAT), aspartate aminotransferase (AST), serum alkaline phosphatase (ALP), alanine aminotransferase (ALT). If no reply to data acquisition received from the author, the approximate statistics were extracted from the chart using Origin software (Fan et al., 2020).

Exclusion criteria

The exclusion criteria were described as follows: (1) Duplicate publication. (2) Headings and abstracts unrelated to MNPs. (3) No data to extract. (4) Lack of relevant animal experiments. (5) Missing specific doses of exposure to MNPs. (6) Meta-analysis, reviews and conference.

Data extraction

After removal of non-eligible articles based on the above criteria, data extraction from the remaining studies was completed by two investigators (Junliang Chen and Ruiquan Qi) and then disagreement

was resolved by a third one (Ye Cheng). The following data were included: first author, year of publication, country, animal species, data (n, mean $\pm$ standard deviation) from experimental and control groups, gender, duration of exposure, dose of exposure, particle size of MNPs, and organs exposed.

Quality assessment

Assessment of the study quality was completed using the Center for Systems Review of Laboratory Animal Experiments (SYRCLE), an animal research adaptation of the Cochrane Risk of Bias Tool (Hooijmans et al., 2014). Evaluation items included sequence generation, baseline characteristics, allocation concealment, random housing, blinding, random outcome assessment, blinding, incomplete outcome data, selective outcome reporting, and other sources of bias.

Statistical analysis

The STATA statistical software version 14.0 was applied to conduct this meta-analysis (Zhang et al., 2018) where high statistical heterogeneity ($P < 0.05$, $I^2 > 50\%$) (Schoenfeld et al., 2019) and random-effects models were applied. Subgroup analysis was performed to determine possible sources of heterogeneity including particle size of MNPs ($< 1 \mu\text{m}$ and $\geq 1 \mu\text{m}$),

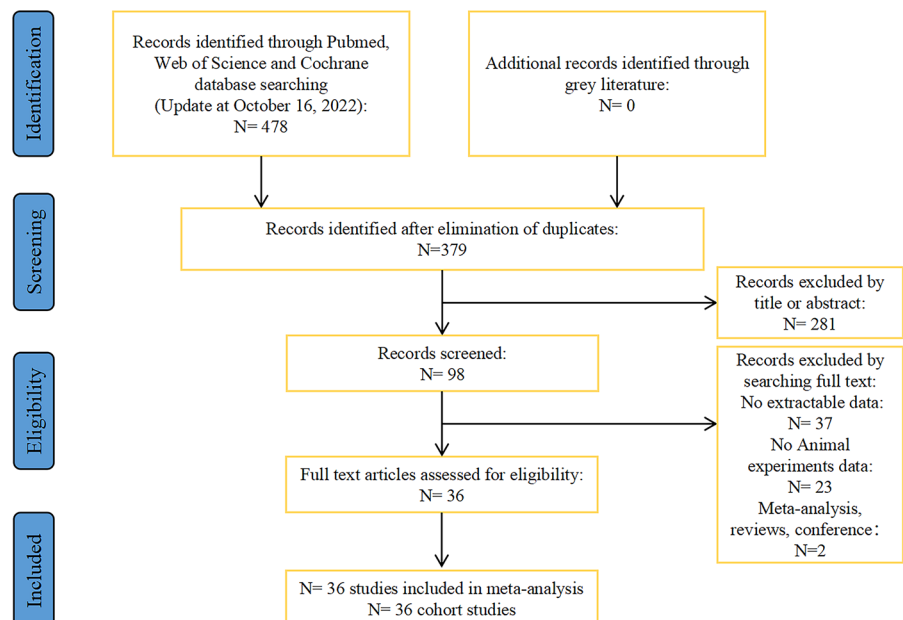
exposure duration (< 30 days, 30–60 days, > 60 days, etc.), exposed organs (liver, kidneys, reproductive organs, etc.), intervention means (orally and gavage), and gender (male and female). Continuous variables were used to assess the standardized mean difference (SMD) with the 95% confidence interval (95% CI) between MNPs treatment group and control. Sensitivity analysis was performed to estimate the influence of individual studies on the final results. Additionally, Egger's test was used to evaluate the publication bias among the included studies (Fan et al., 2020).

Results

Articles exploration and characteristics

A total of 478 articles were identified and applied in the analysis following the above described search strategies in this study, of which 36 articles were validated for the further meta-analysis following the exclusion criteria. The article exclusion was conducted to the ones lacking extractable data ($N=37$), absent of animal experiments data ($N=23$), and being in the form of Meta-analysis, reviews or conference abstracts ($N=2$). The selection process was summarized in Fig. 1. Supplementary Table S2 listed the main features of the meta-analysis included. The samples were collected from various organ sources, such as liver,

Fig. 1 Flowchart of the search strategy. The meta-analysis included rats and mice studies investigating the oxidative effect of MNPs



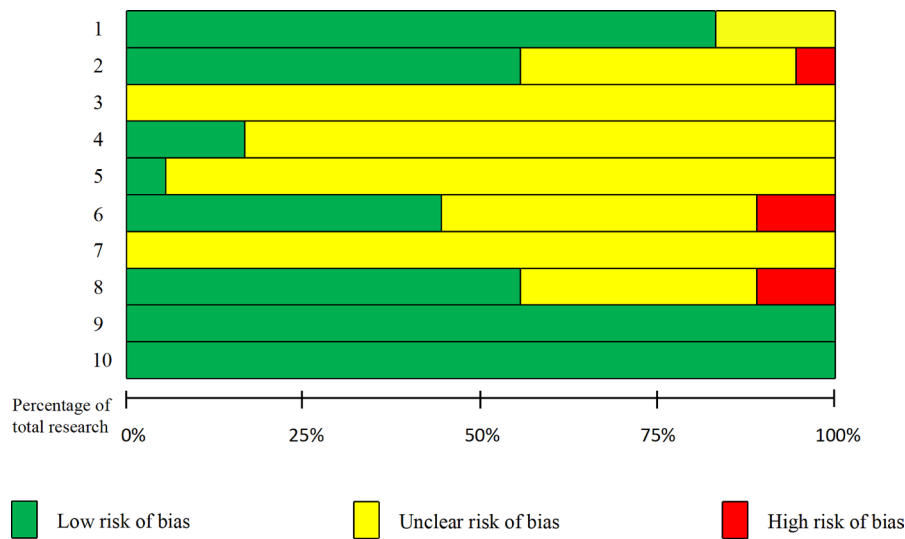


Fig. 2 Risk of bias assessment in the selected articles. (1) Was the allocation sequence adequately generated and applied? (2) Were the groups similar at baseline or were they adjusted for confounders in the analysis? (3) Was the allocation adequately concealed? (4) Were the animals randomly housed during the experiment? (5) Were the caregivers and investigators blinded

to the intervention that each animal received? (6) Were animals selected at random for outcome assessment? (7) Was the outcome assessor blinded? (8) Were incomplete outcome data adequately addressed? (9) Are reports of the study free of selective outcome reporting? (10) Was the study apparently free of other problems that could result in high risk of bias?

kidney, heart, testicle, ovary, lung, colon, duodenum, and the others. The indicators considered in this study include ROS, MDA, CAT, SOD, GPx, GSH, ALT and AST.

Studies quality assessment

The risk of bias assessment for the 36 selected studies was summarized in Fig. 2. “1” represented a sufficiently detailed description of the method used for allocation. “2” was describing all possible prognostic

factors or animal characteristics. “3” represented a sufficient level of detail to describe the methods used for concealing allocation orders. “4” indicated a description of all the measures used to randomly place animals indoors in an animal room. “5” described measures for care providers and researchers involved in blind testing. “6” expressed as whether to randomly select animals for result evaluation. “7” indicated the evaluation measures for the results of the blind method. “8” represented the integrity of the outcome data for each major ending. “9” explained

Table 1 Meta-analysis results for association between MNPs and oxidative stress and serum biochemical parameters

Indicators	Studies (N)	Animals (n)	Heterogeneity		Random effects model	
			I ² (%)	P	SMD(95% CI)	P
ROS	9	164	92.2	<0.001	8.31 (5.52, 11.11)	<0.001
MDA	22	386	90.7	<0.001	3.92 (2.79, 5.06)	<0.001
ALT	11	132	69.1	<0.001	1.01 (0.30, 1.72)	0.005
AST	11	140	83.5	<0.001	1.40 (0.37, 2.43)	0.008
CAT	17	284	94.4	<0.001	−4.68 (−6.55, −2.81)	<0.001
SOD	20	360	94.9	<0.001	−3.93 (−5.68, −2.19)	<0.001
GPx	8	144	96.5	<0.001	−5.62 (−10.08, −1.17)	0.013
GSH	11	188	91.2	<0.001	−3.18 (−4.75, −1.62)	<0.001

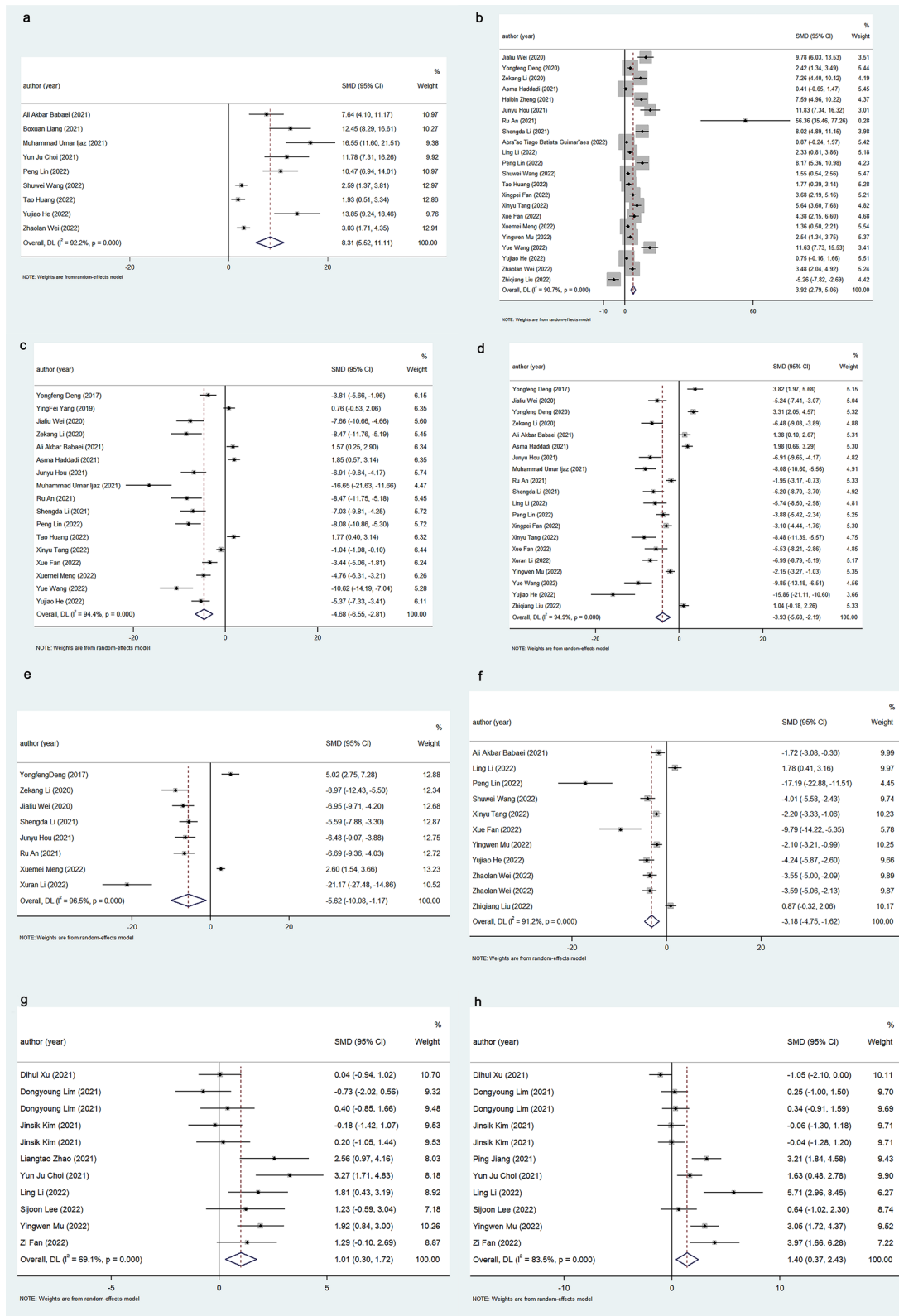


Fig. 3 Forest plots of the effect of MNPs on oxidative stress and serum biochemical parameters. **a** ROS, **b** MDA, **c** CAT, **d** SOD, **e** GPx, **f** GSH, **g** ALT, **h** AST

how to check for selective outcome reports. “10” was among other factors that may contribute to a high bias risk. In terms of bias, most of the included studies were rated as “Low” on the reporting of items including “1”, “2”, “8”, “9” and “10”, but “Unclear” on the rest, few “High” present. Therefore, the overall risk of bias for the quality assessment of these studies was deemed to be low.

Effect of MNPs on oxidative stress and serum biochemical parameters

The random effect model was chosen as a significant heterogeneity ($I^2 > 50\%$) was observed among all the indicators of oxidative stress and serum biochemical parameters, as shown in Table 1 and Fig. 3. The results in Fig. 3 showed that MNPs increased the production of oxidants and blood biochemical indicators, reduced the production of antioxidants, and these indicators exhibited significant heterogeneity ($P < 0.05$, $I^2 > 50\%$). The pooled analysis results suggested distinct effects of MNPs exposure on the expression of different indicators, with increased levels of ROS (SMD=8.31, 95% CI=[5.52, 11.11], $I^2=92.2\%$, $P < 0.001$), MDA (SMD=3.92, 95% CI=[2.79, 5.06], $I^2=90.7\%$, $P < 0.001$), ALT (SMD=1.01, 95% CI=[0.30, 1.72], $I^2=69.1\%$, $P < 0.001$) and AST (SMD=1.40, 95% CI=[0.37, 2.43], $I^2=83.5\%$, $P < 0.001$), and decreased levels of CAT (SMD=-4.68, 95% CI=[-6.55, -2.81], $I^2=94.4\%$, $P < 0.001$), SOD (SMD=-3.93, 95% CI=[-5.68, -2.19], $I^2=94.9\%$, $P < 0.001$), GPx (SMD=-5.62, 95% CI=[-10.08, -1.17], $I^2=96.5\%$, $P < 0.001$) and GSH (SMD=-3.18, 95% CI=[-4.75, -1.62], $I^2=91.2\%$, $P < 0.001$).

The heterogeneity can be influenced by a variety of factors (particle size, days, sample source, etc.). Studies continued to discuss potential heterogeneity in the form of subgroup analysis.

Effects of particle size of MNPs, treatment days, sample source, intervention means and gender on rats/mice oxidative stress indexes.

Subgroup analysis for oxidant and serum indicators of MNPs

The results of subgroup analysis for oxidant and serum indicators of MNPs were shown in Table 2. Specifically, in the subgroup analysis of particle size, the findings indicated that, compared to the treatment group with the larger particle size, the one with smaller particle size exhibited upregulation of ROS, MDA, ALT and AST expression ($P > 0.05$). Given the above results, it could be concluded that MNPs of smaller particle size have a greater impact on various indicators, highlighting the importance of particle size in determining the effects of MNPs (Suppl Fig. S1). Based on the assessment of exposure duration, it was observed that when the treatment time of MNPs exceeded 60 days, there existed a significant increase in the expression level of MDA compared to the other groups ($P < 0.05$), in spite of no notable difference in the effect on the other indicators. This finding suggested a tendency for an enhanced effect on the MDA-related oxidant with increased MNPs exposure time (Suppl Fig. S2). The subgroup analysis for different tissues unveiled tissue sensitivity of ALT and AST, proved by the result that compared to the control group, SMD of ALT and AST were significantly higher in the MNPs-treated liver than that in the other tissues group ($P < 0.05$). Particularly, the heterogeneity of ALT in both the liver group and the serum group decreased to 0.0%, suggesting tissue differences to be the potential source of ALT heterogeneity (Suppl Fig. S3). Non-significant difference was observed between the male and female groups in all oxidant and serum indicators ($P > 0.05$), which indicated that MNPs did not lead to gender-specific differences in oxidant and serum indicators (Suppl Fig. S4). In terms of intervention means, oral administration might remarkably increase the index of MDA than gavage ($P < 0.05$) (Suppl Fig. S5).

Table 2 Subgroup analysis of MNPs on oxidant and serum indicators in rats/mice

Oxidant/ serum indica- tors	Subgroup		SMD	95% CI	<i>P</i>	<i>I</i> ² (%)
ROS	Particle size	< 1 µm	9.50	(4.72, 14.28)	< 0.001	92.2
		≥ 1 µm	6.15	(2.20, 10.10)	< 0.001	93.1
	Exposure duration	< 30 days	10.04	(4.47, 15.62)	< 0.001	93.3
		≥ 30 days	6.45	(2.67, 10.23)	< 0.001	91.9
	Different tissues	Testis	6.09	(1.75, 10.43)	< 0.001	93.5
		Other tissue	9.60	(5.03, 14.16)	< 0.001	91.9
	Intervention means	Orally	7.80	(3.69, 11.92)	< 0.001	93.6
		Gavage	6.83	(0.84, 12.82)	< 0.001	91.4
MDA	Particle size	< 1 µm	4.22	(2.60, 5.83)	< 0.001	91.9
		≥ 1 µm	3.53	(1.94, 5.12)	< 0.001	89.0
	Exposure duration	< 30 days	2.63	(1.62, 3.65)	< 0.001	85.1
		30–60 days	2.51	(− 0.37, 5.39)	< 0.001	93.2
		> 60 days	11.10	(6.48, 15.71)	< 0.001	82.8
	Different tissues	Testis	3.40	(1.72, 5.08)	0.003	78.9
		Liver	3.73	(2.02, 5.43)	0.004	77.7
		Ovary	7.40	(− 0.46, 15.27)	< 0.001	95.8
		Other tissue	5.69	(2.82, 8.56)	< 0.001	92.8
	Different gender	Male	3.83	(2.62, 5.03)	< 0.001	88.7
		Female	4.25	(− 0.40, 8.90)	< 0.001	94.7
	Intervention means	Orally	1.78	(0.43, 3.12)	< 0.001	89.0
		Gavage	8.52	(5.17, 11.87)	< 0.001	91.2
ALT	Particle size	< 1 µm	1.63	(0.62, 2.64)	< 0.001	67.3
		≥ 1 µm	0.36	(− 0.57, 1.28)	0.021	65.3
	Exposure duration	< 28 days	1.16	(− 0.11, 2.44)	0.002	77.0
		≥ 28 days	0.89	(0.00, 1.79)	0.012	65.8
	Different tissues	Liver	2.26	(1.59, 2.94)	0.469	0.0
		Serum	0.06	(0.44, 0.57)	0.624	0.0
	Different gender	Male	0.82	(− 0.06, 1.71)	0.003	69.3
		Female	1.36	(0.03, 2.69)	0.009	74.1
AST	Particle size	< 1 µm	1.93	(− 0.11, 3.98)	< 0.001	88.0
		≥ 1 µm	1.11	(− 0.11, 2.32)	< 0.001	86.1
	Exposure duration	< 28 days	2.02	(0.44, 3.59)	< 0.001	80.3
		≥ 28 days	0.93	(− 0.53, 2.40)	< 0.001	80.7
	Different tissues	Liver	3.10	(1.25, 4.94)	0.018	75.2
		Serum	0.43	(− 0.54, 1.41)	< 0.001	75.5
	Different gender	Male	1.33	(− 0.09, 2.75)	< 0.001	86.5
		Female	1.53	(− 0.15, 3.20)	0.001	81.5

Subgroup analysis for antioxidants indicators of MNPs

Subgroup analysis for antioxidants indicators of MNPs was shown in Table 3. Particle size played a

key role in GSH heterogeneity regulation as shown in Suppl Fig. S1, as the heterogeneity of GSH index *I*² in the ≥ 1 µm group decreased to 42.2% (*P* > 0.05), nevertheless, the expression of GSH and SOD of the ≥ 1 µm particle size group were

Table 3 Subgroup analysis of MNPs on antioxidant indicators in rats/mice

Antioxidant indicators	Subgroup		SMD	95% CI	<i>P</i>	<i>I</i> ² (%)
CAT	Particle size	< 1 µm	−4.61	(−6.72, −2.51)	<0.001	93.6
		≥ 1 µm	−5.09	(−9.44, −0.74)	<0.001	96.0
	Exposure duration	< 30 days	−2.84	(−4.88, −0.80)	<0.001	93.2
		30–60 days	−5.42	(−11.09, 0.25)	<0.001	96.6
		> 60 days	−7.60	(−8.94, −6.26)	0.917	0.0
	Different tissues	Testis	−7.04	(−16.47, 2.39)	<0.001	97.2
		Ovary	−4.40	(−11.66, 2.86)	<0.001	96.5
		Heart	−8.05	(−9.78, −6.32)	0.938	0.0
		Other tissue	−3.06	(−5.12, −1.00)	<0.001	93.1
	Different gender	Male	−4.54	(−6.59, −2.49)	<0.001	94.3
		Female	−4.40	(−11.66, 2.86)	<0.001	96.5
	Intervention means	Orally	−3.84	(−7.56, −0.13)	<0.001	95.8
		Gavage	−5.58	(−9.31, −1.84)	<0.001	95.3
SOD	Particle size	< 1 µm	−4.90	(−6.77, −3.04)	<0.001	92.7
		≥ 1 µm	−1.62	(−5.09, 1.85)	<0.001	96.4
	Exposure duration	< 30 days	−4.17	(−6.65, −1.70)	<0.001	94.7
		30–60 days	−2.16	(−5.82, 1.50)	<0.001	96.1
		> 60 days	−5.20	(−7.49, −2.91)	<0.001	82.6
	Different tissues	Liver	−1.72	(−4.91, 1.47)	<0.001	93.6
		Heart	−4.92	(−6.40, −3.44)	0.208	36.3
		Ovary	−1.24	(−4.07, 1.60)	<0.001	93.4
		Testis	−3.60	(−11.53, 4.33)	<0.001	97.8
	Different gender	Male	−4.71	(−7.28, −2.13)	<0.001	95.8
		Female	−2.08	(−4.86, 0.70)	<0.001	93.3
	Intervention means	Orally	−1.46	(−4.22, 1.31)	<0.001	96.2
		Gavage	−4.50	(−7.07, −1.93)	<0.001	92.8
GPx	Particle size	< 1 µm	−7.83	(−17.89, 2.22)	<0.001	97.7
		≥ 1 µm	−4.58	(−9.79, 0.63)	<0.001	94.9
	Exposure duration	< 30 days	−3.43	(−10.98, 4.12)	<0.001	96.6
		≥ 30 days	−6.67	(−7.87, −5.47)	0.626	0.0
	Different gender	Male	−5.02	(−11.04, 1.00)	<0.001	97.0
		Female	−6.70	(−8.24, −5.16)	0.970	0.0
	Intervention means	Orally	−3.43	(−10.98, 4.12)	<0.001	96.6
		Gavage	−6.67	(−7.87, −5.47)	0.626	0.0
GSH	Particle size	< 1 µm	−3.45	(−5.90, −1.01)	<0.001	93.4
		≥ 1 µm	−3.20	(−4.11, −2.29)	0.159	42.2
	Exposure duration	< 30 days	−4.23	(−6.57, −1.89)	<0.001	92.6
		≥ 30 days	−1.97	(−4.14, 0.20)	<0.001	90.1

slightly upregulated ($P > 0.05$), when compared to the < 1 µm particle size group. As presented in Suppl Fig. S2, CAT was proved to be more sensitive to the exposure duration of MNPs with marked SMD reduction in the expression level of CAT in

the group of > 60 days, compared to that in the group of < 30 days ($P < 0.05$). Moreover, heterogeneity of CAT and GPx were observed downregulated, as I^2 of both indicators decreased to 0.0% ($P > 0.05$) when exposure time was ≥ 30 days (for

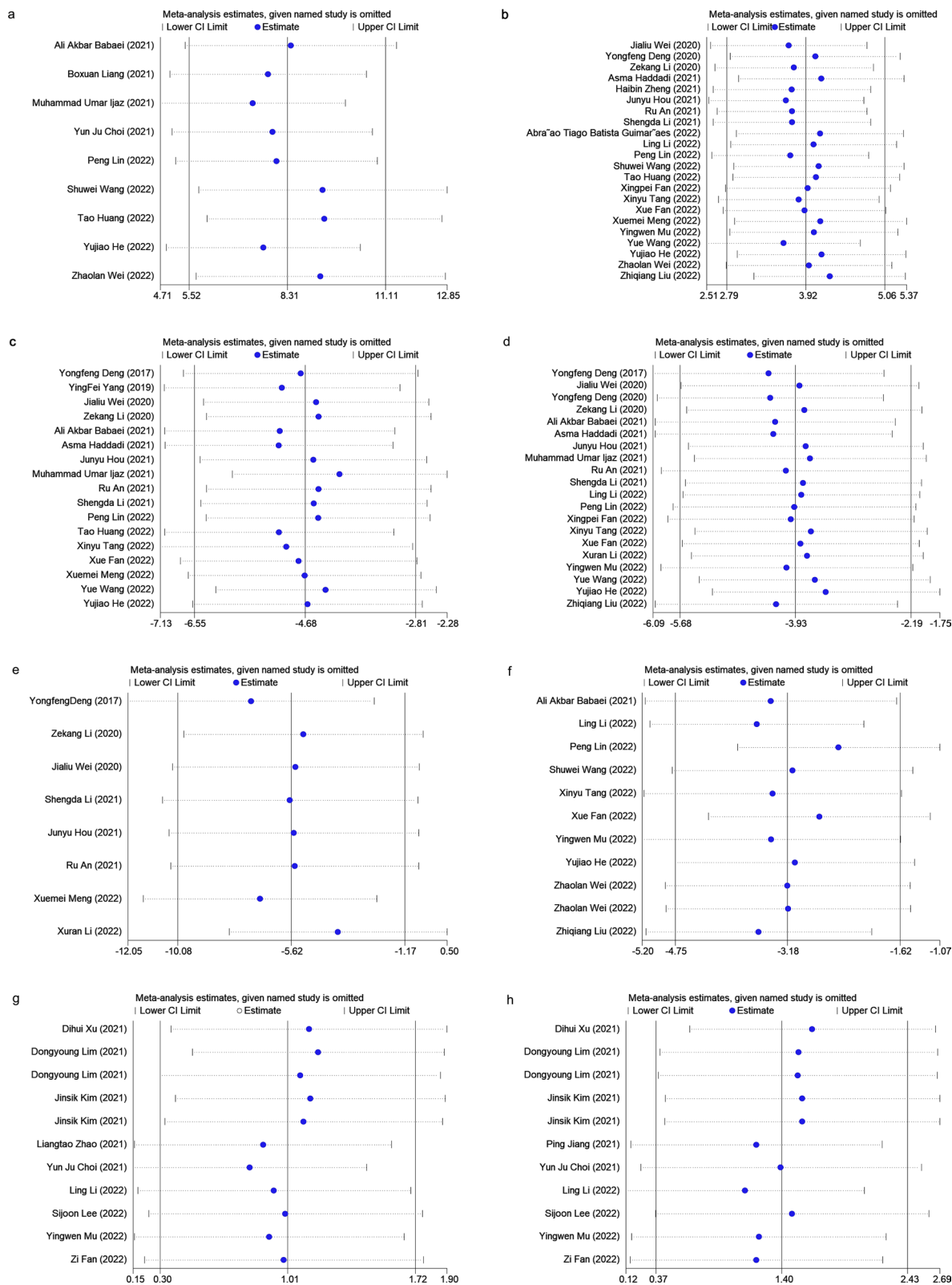


Fig. 4 A sensitivity analysis for data of different oxidative stress and serum indicators. **a** ROS, **b** MDA, **c** CAT, **d** SOD, **e** GPx, **f** GSH, **g** ALT, **h** AST

GPx) and > 60 days (for CAT) respectively. The impact of tissue types was primarily on CAT and SOD heterogeneity as shown in Suppl Fig. S3. For CAT, I^2 decreased to 0.0% ($P > 0.05$) in the heart group, while for SOD, I^2 decreased to 36.3% ($P > 0.05$). Similar to oxidant and serum indicators, no significant change in antioxidant indexes between male and female groups ($P > 0.05$) was marked as in Suppl Fig. S4, indicating limited gender-specific effect of MNPs on antioxidant indexes. Little difference in both indicator expression and heterogeneity with each intervention means was seen in Suppl Fig. S5, except GPx, whose heterogeneity decreased to 0.0% ($P > 0.05$) in the gavage group.

Sensitivity analysis and publication bias

Sensitivity analysis were performed to assess the stability of the eligible studies (Zhang et al., 2022a, b). The results showed that when any study was removed from the model, the remaining studies showed values within the 95% confidence interval. The overall results for oxidative stress and blood biochemical indicators in MNPs remained stable. Therefore, the results of sensitivity analysis indicated that the included studies were not significantly biased by individual studies, as shown in Fig. 4.

The results in Fig. 5 illuminated that, except for ALT, the Egger's test of the remaining indicators included in the study exhibited asymmetrical distribution. And the Egger's test was conducted to detect the P values of each indicator, ROS ($P < 0.001$), MDA ($P < 0.001$), CAT ($P < 0.001$), SOD ($P < 0.001$), GPx ($P = 0.009$), GSH ($P = 0.023$), ALT ($P = 0.154$), AST ($P = 0.025$). Therefore the trim and fill test were applied to further analysis. Based on the results, there were 3 studies added for MDA (SMD = 3.25, CI = [2.07–4.44]) and 1 study added for AST (SMD = 1.11, CI = [0.06–2.17]). No missing study was identified in the results of ROS, CAT, SOD, GPx and GSH.

Discussion

MNPs, recognized as a detrimental environmental pollutant, have been detected in various human tissues, including feces (Schwabl et al., 2019), blood (Leslie et al., 2022) and lungs (Jenner et al., 2022). Numerous studies have illuminated the adverse impact of MNPs, such as ovarian dysfunction in rats (Babaei et al., 2022), testicular tissue damage (Zhao et al., 2021), and reproductive fertility disruption in mice (Wei et al., 2022). Furthermore, MNPs have been found to significantly impair lung function, break down the integrity of the heart muscle (Krishnan, 2023), and even facilitate the progression of cancer (Xiao et al., 2023), among others.

Studies have shown that MNPs induced lung injury by triggering oxidative stress pathways (Zhang et al., 2022a, b). The MNP-induced oxidative stress response can lead to cell damage, increased vulnerability to neuronal disease, and brain malfunction (Prüst et al., 2020). Moreover, MNPs with different surface functionalizations, have been observed to induce severe oxidative stress in HepG2 cells, potentially affecting liver function (He et al., 2020). In addition, MNPs are capable to disperse and aggregate in the circulatory system, posing a threat to heart function and causing toxicity to vascular sites (Zhu et al., 2022). Furthermore, the toxicity of MNPs might be related to particle size, exposure duration, intervention methods, and specific tissues involved (Dang et al., 2022). Therefore, our study aimed to provide insights into the relationship between MNPs and oxidative stress as well as serum biochemical parameters. In the study of An et al. (2020), the effects of nano-titanium dioxide exposure on rats and mice were investigated with subgroup analysis on potential toxicity factors, including different intervention pathways, exposure time and sample sources, using oxidative stress as indicators.

Although the specificity of each oxidative stress and serum biochemical parameters indexes may vary, it's noteworthy that following MNPs treatment, certain indices including ROS, MDA, AST, and ALT ($P < 0.05$), exhibited a substantial increase, while others, such as SOD, GSH, GPx, and CAT ($P < 0.05$), showed a significant decrease. These findings aligned with the results of a meta-analysis conducted in the meta-analysis by Fan et al. (2020), which demonstrated a significant elevation in the levels of oxidants

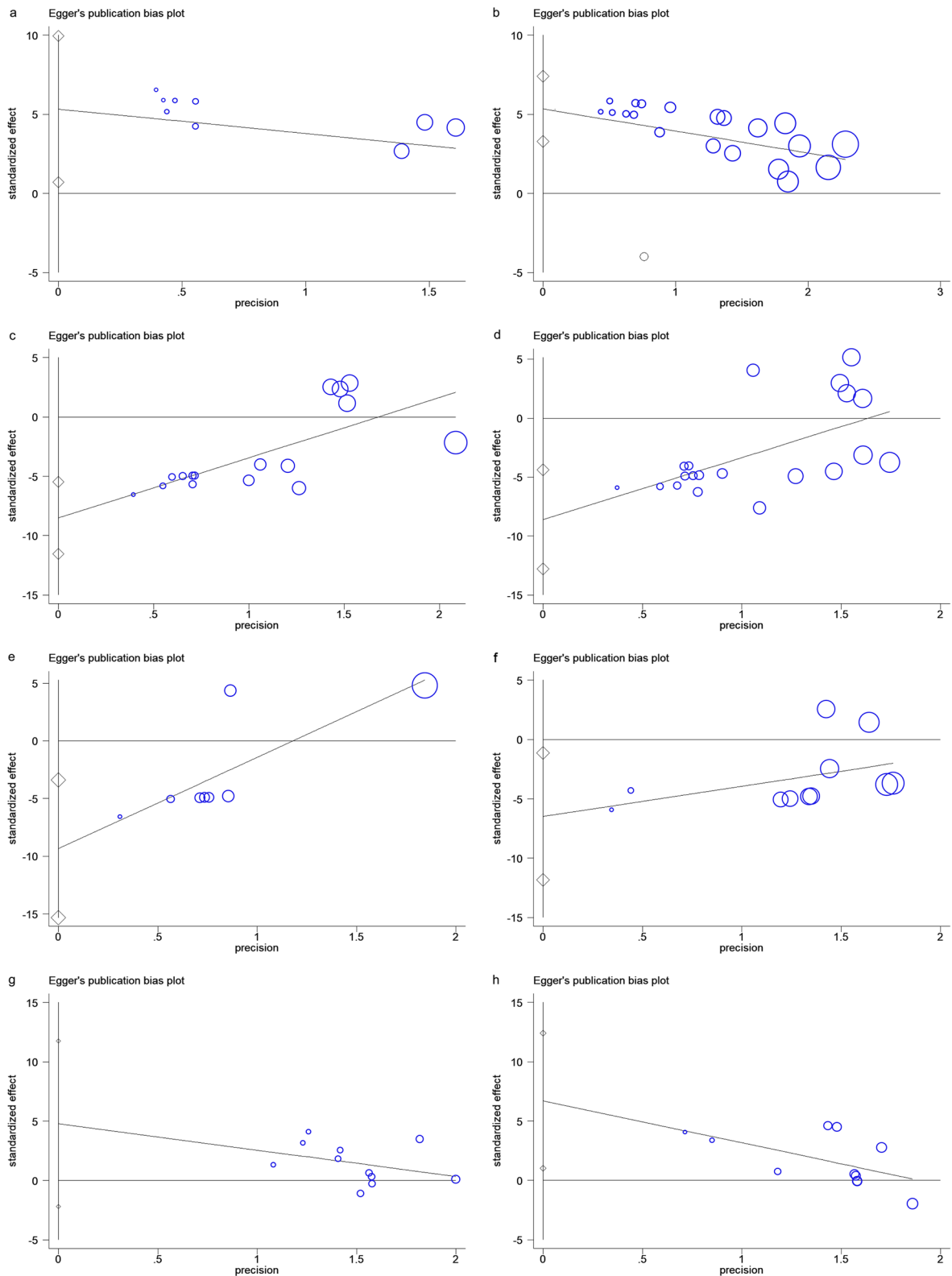


Fig. 5 Egger's test for data of different oxidative stress and serum biochemical parameters. **a** ROS, **b** MDA, **c** CAT, **d** SOD, **e** GPx, **f** GSH, **g** ALT, **h** AST

like MDA, GSOG, ROS, and H_2O_2 ($P < 0.05$) and a marked reduction in the levels of antioxidants such as CAT, GPx, GSH, and SOD ($P < 0.05$), which reinforced the validity of our results.

Exposure to microplastics encompassed a diverse range of factors, including plastic size variation, polymer compositions, surface modifications, exposure pathways, and duration of exposure (Lett et al., 2021). Consequently, subsequent studies suggested that the extent of oxidative damage and serum biochemical parameters may be affected by the particle size, target organ, exposure duration, gender, and intervention of MNPs. Subgroup analysis based on particle size indicated that MNPs with smaller particle size led to slight increase in ROS, MDA, ALT and AST expression. Simultaneously, the expression of SOD, GPx, and GSH were downregulated ($P > 0.05$). These findings suggested that exposure to both large and small particle size MNPs can impact the oxidation system homeostasis in rats/mice, however, the smaller the particle size, the greater the effect on oxidation indexes. In the related study by Banerjee et al. (2022), it was noted that smaller sized particles exhibited more susceptibility to internalization by HepG2 cells than larger particles, and were found to be more detrimental to cell viability with rising concentration.

The results of the subgroup analysis based on exposure duration revealed that prolonged exposure to MNPs (> 60 days) resulted in a significant increase in MDA levels ($P < 0.05$) but a notable decrease in CAT levels ($P < 0.05$). It suggested that extended MNPs exposure may lead to an accumulation of oxidizing substances, posing adverse health risks. Xu et al. (2019) emphasized the critical role of exposure duration in the effects of nanoplastics on lung epithelial cells, with a more pronounced impact observed as exposure time extended. Furthermore, there was no significant indicator difference observed between male and female groups ($P > 0.05$), and no marked variations in the levels of ROS, CAT, SOD and GPx when comparing the oral group and the gavage group ($P > 0.05$). However, it's worth noting that oral intervention can lead to a significant increase in MDA ($P < 0.05$). This suggested that these interventions might exert distinct effects on certain specific

indicators while leaving most markers unaffected. The study conducted by Zhang et al. (2022a, b), presented no significant disparities in the levels of MDA, CAT, GPx and GSH between the oral group and the gavage group ($P > 0.05$), while oral intervention resulted in a significant reduction in SOD ($P < 0.05$). The above results collectively underscored that oral administration might impact specific indicators differently.

The results revealed that the absolute SMD values of ALT and AST in the liver of the MNPs-treated group were significantly higher than those of other tissues versus the control group ($P < 0.05$). In the case of CAT, heterogeneity decreased to 0.0% in the heart group, and in ALT, heterogeneity decreased to 0.0% in both the liver and serum groups. The findings showed that MNPs had a more obvious effect on ALT and AST in liver tissues, and the heart appeared to be the primary source of heterogeneous factor in CAT, and liver and serum were the main heterogeneous factors in ALT. Some studies have suggested that MNPs tend to accumulate in the liver through absorption or epidermal infiltration through the blood circulation, affecting the liver function and causing subsequent damage to other organs (Haldar et al., 2023). At the same time, ALT and AST have recently gained recognition as risk factors for cardiovascular disease (CVD) (Choi et al., 2018). Lee et al. (2016) reported that elevated ALT and AST levels were associated with increased CVD mortality and all-cause mortality. Overall, the analysis illustrated the potential impact of elevated levels of these indicators on CVD.

Due to the heterogeneity of the studies, we analyzed the results further using a random-effects model for a better understanding of the results. We performed subgroup analysis of various possible influencing factors to pinpoint sources of heterogeneity. We figured out that heterogeneity might be determined by particle size, time of exposure, tissue types, gender difference, and intervention methods. These findings were consistent with those in the study of Zhang et al. (2022a, b), where heterogeneity was attributed to variations in bisphenol A dosage, duration of exposure, and specific rats/mice tissues.

This study faced several limitation factors that may have affected the accuracy. Firstly, the eight indexes of ROS, MDA, CAT, SOD, GPx, GSH, ALT and AST were assessed by funnel analysis and the restricted the number of included studies could influence the outcomes. Secondly, despite the observed

heterogeneity in particle size, tissue types, exposure duration, gender difference and MNPs intervention modalities, multiple additional factors, such as the frequency of MNPs treatment, experimental designs, testing methods and so on, could also contribute to these variations. In addition, only English articles were included in the meta-analysis of MNPs, potentially omitting valuable research published in other languages. Furthermore, the study focused on rats and mice excluding investigations involving other marine animals or organisms which could have further elucidated the relationship between MNPs and oxidative damage. As such, subsequent research should encompass a broader range to provide more robust evidence for understanding the mechanisms underlying the toxicity of MNPs.

Conclusion

The meta-analysis of MNPs in this study emphasized the fact that MNPs could increase oxidant levels, elevate serum biochemical parameters and decrease antioxidant levels. Further research found that smaller MNPs particle size, longer exposure time, and oral administration might cause the imbalance of the oxidative antioxidant system and elevate serum biochemical parameters, leading to potential hazards in liver system. The results of this meta-analysis provided reliable experimental data for exploring the oxidative stress and CVD caused by MNPs, which help to found a reliable basis for the toxicity mechanism of MNPs, and the prevention of MNPs-induced oxidative stress and CVD.

Author contributions JLC extracted the statistics and wrote original manuscript. RQQ identified eligible studies, and checked the manuscript. YC checked and managed the data. LW conducted software analysis. XYZ was responsible for conceptualization and fund acquisition. All authors have read and agreed to the published version of the manuscript.

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Data availability All data generated and analyzed during this study are included in this published article.

Declarations

Competing interests The authors declare no competing interests.

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